

Supam Roy

+91 6291700361 | supamroy328@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

National Institute of Technology Durgapur
Bachelor of Technology in Computer Science and Engineering

Durgapur, West Bengal
June 2022 – May 2026

EXPERIENCE

Research Intern

January 2026 – April 2026

IISC Bangalore

Bangalore, India

- Led a **4-member team** in architecting a real-time **MITM engine** using **ARP poisoning** and **Linux NetfilterQueue** to intercept and forge live **IEC 61850** traffic, preserving stateful TCP sessions while processing **1,000 packets/sec**.
- Designed a **multithreaded packet-processing pipeline** with **Redis TimeSeries** for live stream buffering and historical payload retrieval, enabling stateful out-of-band packet manipulation with less than **0.5ms lookup latency**.
- Developed a memory-efficient **PyQt5 packet analysis application** with streaming PCAP processing, enabling analysis and manipulation of **300+ MB network captures** without GUI freezes or excessive memory consumption.

Software Engineer Intern

May 2025 – July 2025

Tailored AI

Bangalore, India

- Engineered an end-to-end **Document Copilot** using **FastAPI, React, and PostgreSQL**, automating SEC 10-K/10-Q research workflows and saving analysts an estimated **3+ hours per week**.
- Optimized **SEC filing ingestion** by eliminating I/O bottlenecks with **asyncio-driven concurrent batching**, reducing embedding latency by **30%**.

PROJECTS

PropSphere | *PostgreSQL, Redis, RabbitMQ, gRPC, Docker, AWS ECS*

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- Architected a **5-microservice event-driven platform** using **Saga orchestration** and a **Transactional Outbox** over **RabbitMQ**, ensuring at-least-once delivery with idempotent processing and eliminating dual-write inconsistencies
- Built concurrency-safe billing by combining **Redis mutexes** with atomic **PostgreSQL transactions**, preventing duplicate wallet deductions under **500 concurrent users** and **5,500+ contention attempts**
- Automated **CI/CD with GitHub Actions and Terraform** for **AWS ECS Fargate** deployments and shrank Docker images by **63%** using multi-stage Alpine builds, enabling rolling deployments

ThrottleX | *Node.js, Express, TypeScript, Redis, Lua, Jest*

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- Built a **distributed rate limiting engine** exposing **6 production-grade algorithms** behind a unified **RateLimiter interface**, letting services swap strategies via config with no code changes
- Eliminated **TOCTOU race conditions** across horizontally scaled instances by executing all state transitions as atomic **Redis Lua scripts** in a single round-trip
- Validated correctness under load with **113 automated tests** covering concurrency and failure scenarios, benchmarking up to **190K+ requests/sec** at **0.21ms P99 latency**

Synapse Social | *Next.js, React, PostgreSQL, Inngest, OAuth*

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- Shipped an **event-driven social publishing platform** on **Next.js 15 and Inngest**, orchestrating background jobs to handle OAuth token refresh, fault-tolerant media uploads, and scheduled posting across **8 platforms**
- Enforced tenant isolation by mapping **Clerk JWT claims** to **PostgreSQL Row-Level Security** policies, guaranteeing database-level separation of user channels, ideas, and scheduled posts

ACHIEVEMENTS & CERTIFICATIONS

- Solved **1,000+ algorithmic problems** on [LeetCode](#), strengthening expertise in data structures and algorithms
- Achieved a **LeetCode Knight** rating of **1862**, placing in the **Top 5.72%** globally across **18** rated contests
- Earned the **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** certification

TECHNICAL SKILLS

Languages: C++, TypeScript, JavaScript, Python

Frontend: React, Next.js, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, WebSockets, gRPC

Data & Messaging: PostgreSQL, MongoDB, Redis, RabbitMQ

Cloud & DevOps: AWS (EC2, S3), Docker, CI/CD

Core CS: DSA, System Design, DBMS, Computer Networks